CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY

College of Computer Studies

Bachelor of Science in Information Technology

IT411 CAPSTONE 2 DELIVERABLE — STANDALONE EXTRACTION

PlayIT — Software Requirements Inventory

Complete Inventory of Specific Requirements (FR-01 to FR-14 & NFR-AUD-01 to NFR-SEC-01)

Course	IT411 (Section G7) — Capstone & Research 2 | Semester 1, AY 2026–2027
Document Type	Standalone Deliverable: Requirements Inventory
Master Source	PlayIT Software Requirements Specification (SRS v3.0, revision 3.8)
Extracted Scope	Section 3: Specific Requirements (Pages 9–15 of SRS v3.0)
Publication Date	September 26, 2026 (Revised October 14, 2026)
Document Status	Submitted for Adviser Review and Evaluation
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3. Specific Requirements

3.1 External Interface Requirements

3.1.1 Hardware Interfaces

- **Microphone Interface:** Standard built-in device microphone supporting 16,000 Hz, 16-bit linear PCM mono input.
- **Speaker Interface:** Internal device audio transducer supporting 44.1 kHz, 16-bit uncompressed WAV / OGG output.

3.1.2 Software Interfaces

- **Operating System:** Android API Level 26+ (`minSdkVersion = 26`, `targetSdkVersion = 34`).
- **Offline ASR Engine:** Vosk Android SDK v0.3.47 bundled with a lightweight acoustic model.
- **Persistence Engine:** Android Jetpack Room 2.6+ backed by SQLite.
- **UI Toolkit:** Jetpack Compose 1.5+ utilizing Material Design 3 tokens.

3.1.3 Communications Interfaces

- No runtime external network communication interfaces are utilized. All data processing occurs locally.

3.2 Refactored Functional Requirements (FR)

Module 1: Letter-Sound Acquisition ("Hear It")

[FR-01] Carrier Narration & Audio Guidance

- **Description:** The system shall play clear, accent-neutral spoken carrier instructions voiced by the pedagogical persona (Kokoro-82M, voice mix `af_heart` 0.7 + `af_bella` 0.3).
- **Pre-condition:** User selects an unlocked letter node and enters Hear It.
- **Main Flow:**
  1. System initializes the audio subsystem.
  2. System loads the carrier audio asset (`car_listen.wav`) and outputs audio through the device speaker within ≤ 0.2 s.
  3. Mascot animates speech synchronized with audio playback.
- **Post-condition:** Carrier instruction completes, transitioning to the letter reveal.

[FR-02] Pure Phoneme Modeling & Modeling Sequence (Refactored · P0)

- **Validation Finding:** MVP teachers flagged vocal schwa intrusions (e.g., /m/ pronounced as "ma, ma, ma" rather than pure continuous /m/ [m:]), causing blending interference in CVC synthesis (PED-08).
- **Requirement:**
  1. The system shall play an acoustically pure phoneme model for each of the 26 letters. Continuous phonemes (/a, e, i, o, u, f, l, m, n, r, s, v, z/) shall be sustained for approximately 800 ms. Stop consonants (/b, c, d, g, h, j, k, p, q, t, w, x, y/) shall be clipped cleanly at ≤ 250 ms (≤ 300 ms for breaths/glides).
  2. Under no circumstance shall any phoneme audio contain an added trailing schwa (/ə/), syllabic vowel, or letter-name pronunciation.
  3. Phoneme audio clips shall be stored and played independently from carrier phrases and example-word audio.

4. The modeling sequence shall execute runtime assembly:
    - Carrier (*"Listen!"*)
    - Visual Letter Reveal with Picture Mnemonic
    - Carrier (*"This letter says..."*)
    - Pure Phoneme (3 iterations with 500 ms pauses). **Amended (team decision 2026-10-11, card 28c):** no mouth-shape cue is shown during the sequence, so the child watches the letter and listens; the cue supports Say It (FR-03.4, NFR-ACC-01).
    - Key Word Audio (e.g., *"mouse"*)
    - Pure Phoneme (1 iteration)
    - Carrier (*"Great listening! Tap the big button."*), which points to the Next button. Hear It is listen-only, so it no longer ends with *"Say it with me!"* (team decision 2026-10-10, card 28).
5. **[Planned — Round 2 Scope]** The entire modeling sequence shall execute within ≤ 15 s.
6. A persistent Audio Replay ("Ear") button shall be present on screen to allow immediate replay of steps 3–6 at any time.

- **Acceptance Criteria:** 26 of 26 phoneme clips rated "Pure" by at least 3 of 4 DepEd reading teachers (HI-1); modeling sequence completed within ≤ 15 s (HI-2).

Module 2: Speech Production & Formative Tutoring ("Say It")

[FR-03] Tutoring Loop, Active Microphone States & Prompt Ladder (Refactored · P1)

- **Validation Finding:** Facilitator notes recorded MVP learners hesitating at a mic button that gave no sign of listening, and perceived ease was the lowest Smileyometer item (7 of 16 children did not choose the top face). The adviser's refactor directive asks for formative correction without penalties.
- **Requirement:**
    1. The Say It module shall transition through a structured tutoring sequence:
        - **We Do Step:** Two un-scored choral practice turns (*"Say it with me!"*) with animated mascot lip movements.
        - **You Do Step:** Independent vocal turn. **[Planned — Round 2 Scope]** The microphone shall activate automatically following the carrier cue *"Your turn!"* (or upon explicit tap of the microphone button).
    2. The microphone component shall explicitly display one of four mutually exclusive operational states, each with its own colour, icon, motion and text label, and none of them red (team decision 2026-10-06: Heard takes the place of a separate Processing spinner, and the ripple stays time-based until the `AudioRecord` loop of SDD §3.2 lands after Round 2):
        - **IDLE:** Green microphone, stationary ("Tap and say it").
        - **LISTENING:** Gold microphone with an animated ripple ("I'm listening..."). Target design: the ripple's radius scales with live input RMS amplitude (0.0–1.0).
        - **HEARD:** Light-lavender microphone with bouncing dots as soon as speech is detected, while Vosk finishes the result ("I hear you!").
        - **RESULT:** Green check with a success chime ("Yes!"), or soft orange with a formative guidance prompt ("Let's try again").
    3. **Visual Latency:** Tap-to-Listening transition shall occur within ≤ 100 ms.
    4. **Formative Prompt Ladder:** If vocalization is absent after 3 s or an error is detected, the system shall execute progressive pedagogical scaffolding across up to three scored attempts:
        - *Attempt 1 Miss:* Specific spoken correction addressing error type (e.g., Letter Name: *"That's the letter's name. Its sound is /m/."*; Added Vowel: *"Almost! Just /m/, no 'ah'."*) + re-model + *"Your turn."*

    - *Attempt 2 Miss:* Slower re-model with enlarged visual mouth articulation cue + *"Your turn."*
    - *Attempt 3 Miss:* System initiates together-practice (*"Let's say it together: /m/."*), marks the letter as `NEEDS_PRACTICE` in Room DB, advances the lesson without failure screens, and schedules a spaced recall recheck.
  5. **Heart Protection:** The Say It module shall **never deplete player hearts** upon incorrect pronunciation or recognition rejection.
  6. **Scoring Mode (hybrid; team decision 2026-10-01):** The scored Say It check uses **word mode**: the child says the key word (e.g., "mouse"), and the judge matches it against a per-letter grammar of the key word plus that letter's foils (letter name, added vowel). Pure-sound scoring is used for the recall check only if the on-device Vosk test with children shows that held letter names are rejected; the Week 4 spike found that Vosk cannot confirm a correct pure sound (`docs/spikes/vosk-foil-spike.md`), so pure-sound scoring is currently disabled. Letter names and added vowels are never accepted.
- **Acceptance Criteria:** Tap-to-Listening latency ≤ 100 ms (TC-MIC-01); all 3 prompt ladder branches successfully execute (TC-SAY-01..04); 0 hearts deducted across all Say It errors.

[FR-NEW-REC] Spaced Retrieval Warm-Up and Recall Mastery Checks (New · P1)

- **Requirement:**
  1. Phonetic mastery shall be determined by retrieval from memory rather than immediate imitation.
  2. Each learning session shall begin with a 2–3 letter **Warm-Up Retrieval Check** presenting learned letters in isolation (no audio model provided).
  3. Priority queue logic: Letters flagged as `NEEDS_PRACTICE` shall appear first, followed by oldest `SECURE` letters.
  4. Each completed node shall culminate in an **End-of-Session Recall Check**.
  5. A letter shall achieve permanent `SECURE` mastery status only after producing a correct vocalization during an unassisted recall check.
- **Acceptance Criteria:** TC-REV-01 to TC-REV-03 pass; **[Planned — Round 2 Scope]** ≥ 70% of previously un-mastered letters correctly produced during end-of-session recall checks (SI-4).

Module 3: Phoneme Discrimination ("Find It")

[FR-04] Dynamic Grid Discrimination

- **Requirement:** The system shall display a dynamic 3 × 2 grid containing exactly 3 target images matching the current phoneme sound and 2 distractor images selected strictly from previously mastered letters.
- **Pass Rule:** The learner must identify all 3 target items before hearts reach 0.
- **Accuracy Metric:** Discrimination Accuracy = Correct Taps / Total Taps × 100%. Passing target: ≥ 80%.

[FR-05] Heart Depletion & Buffer in Find It

- **Requirement:** The learner begins Find It with 5 hearts. Each incorrect card tap deducts exactly 1 heart and plays a gentle corrective acoustic tone. If hearts reach 0, the activity pauses and reloads with a 3-heart retry buffer.

Module 4: Word Construction Checkpoint ("Blend It")

[FR-13] Decodable CVC Word Bank & Synthesis Engine (Refactored · P1)

- **Validation Finding:** Post-validation pedagogical audit revealed 5 words in the MVP word bank that contained unintroduced digraphs and vowel teams (e.g., *ai* in AIM, *ee* in BEE, *oy* in TOY/BOY, *oo* in ZOO), violating phonics decodability rules.

- **Requirement:**
  1. Every Blend It word shall be 100% decodable using exclusively the letter-sounds unlocked in the current and preceding chapters.
  2. The seeded database shall replace the 5 invalid words:
     - Chapter 1: Remove **AIM** → Replace with **AM** (Letters: *a, m*)
     - Chapter 2: Remove **BEE** → Replace with **SUM** (Letters: *s, u, m*)
     - Chapter 3: Remove **TOY** → Replace with **TUB** (Letters: *t, u, b*)
     - Chapter 3: Remove **BOY** → Replace with **YAM** (Letters: *y, a, m*)
     - Chapter 7: Remove **ZOO** → Replace with **ZIP** (Letters: *z, i, p*)
     - Chapter 7: Retain **QUIZ** strictly as a documented pedagogical exception flagged with `isDocumentedException = true` (explaining *qu* = /kw/).
  3. The activity shall display 3 letter slot boxes and draggable/tappable letter tiles. Learner must sequence phonemes into valid words.
- **Acceptance Criteria:** 100% of seeded word bank validated as decodable against unlocked chapter letters (TC-BLD-01).

Module 5: Mastery Progression, Gamification & Multi-Profile

[FR-06] Star Calculation Rules

- **Requirement:** Upon node completion, stars shall be awarded:
  - **3 Stars:** 100% discrimination accuracy and 0 hearts lost.
  - **2 Stars:** ≥ 80% accuracy with ≤ 2 hearts lost.
  - **1 Star:** Completed activity, but with > 2 hearts lost or accuracy between 50%–79%.

[FR-07] Heart Recovery System

- **Requirement:** The system shall award +1 heart for every 3 consecutive correct taps across Find It. Recovered hearts shall be capped at the session starting pool (5 for fresh sessions, 3 for restart sessions).

[FR-08] Linear Marungko Sequence Enforcer

- **Requirement:** Letter node N + 1 shall remain strictly locked until Letter node N has achieved at least 1 star across all three sub-levels.

[FR-09] Daily Practice Streaks

- **Requirement:** The system shall increment the active profile's streak counter upon the completion of at least one learning node within a calendar day. Streak reset occurs only after 24 hours of inactivity without penalizing stars or unlocked levels.

[FR-14] Multi-Profile Management for Shared Stations (Refactored · P1)

- **Validation Finding:** All 4 teachers judged PlayIT ready as a supplementary tool for Grade 1 classes (PED-12). Classroom use means shared tablets, so multi-profile support, which the MVP already had, is formalized as a requirement.
- **Requirement:**
  1. The application shall support up to 6 distinct child profiles on a single device without requiring online authentication.
  2. Each profile shall possess fully isolated progression states, star tallies, heart pools, unlocked biomes, and telemetry event logs.
  3. An intuitive, visual Avatar Switcher shall be accessible from the main splash screen.
  4. Actions executed under Profile A shall never modify or corrupt data belonging to Profile B.

- **Acceptance Criteria:** Unit/Instrumented test TC-DB-03 confirms complete profile isolation across concurrent SQLite write operations.

Module 6: Telemetry, Parent Dashboard & Diagnostics

[FR-10] Offline Parent Dashboard Display

- **Requirement:** An offline analytics dashboard shall render learning metrics: accuracy trend per letter, total attempts, hearts lost, time-on-task, retention scores, and color-coded flags (Green: ≥ 80% mastered, Yellow: 50–79% developing, Red: < 50% at-risk).
- **Access Control:** Protected by a lightweight pediatric arithmetic gate (e.g., *"Solve 7 + 5 = ?"*) to prevent accidental child access.

[FR-12] Local PDF Progress Report Generation

- **Requirement:** The system shall compile the active profile's performance analytics into a multi-page, formatted PDF document written to local app-specific storage using Android's native `PdfDocument` API.

[FR-NEW-TEL] Offline Telemetry Logging & Local CSV Export (New · P1)

- **Validation Finding:** Round 1 could not measure speech latency, agreement with teachers or error types because the MVP logged no interaction events; teachers asked for a way to track learners' progress (Weeks 1–2 validation summary).
- **Requirement:**
  1. The system shall automatically record structured interaction telemetry to a local Room table (`TelemetryEvent`) upon every user event.
  2. Logged events shall capture: `profileId`, `sessionId`, `chapter`, `letter`, `module`, `eventType`, `elapsedRealtimeMs`, `wallClockEpoch`, `resultValue`, and `asrConfidence`.
  3. Event types: `session_start`, `session_end`, `node_enter`, `node_complete`, `hear_play`, `mic_tap`, `speech_end`, `asr_result`, `feedback_shown`, `find_tap`, `find_feedback_shown`, `blend_submit`, `heart_change`, `idle_reprompt`.
  4. The Parent Dashboard shall provide a PIN-protected **Export Telemetry (CSV)** function that compiles raw event logs into standard comma-separated text files shared via the native Android Share Sheet without network calls.
- **Acceptance Criteria:** TC-TEL-01 verifies schema fields; TC-EXP-01 confirms PIN lock; TC-EXP-02 confirms zero network transmission.

3.3 Non-Functional Requirements (NFR)

3.3.1 Acoustic Quality & Speech Recognition Performance

[NFR-AUD-01] Phoneme Audio Production & Three-Stage Release Gate (New · P0)

- **Requirement:**
  1. Audio assets shall be generated using the Kokoro-82M neural model (Apache-2.0). Held (continuous) sounds may instead use Chatterbox-Turbo (MIT), voice-cloned from a Kokoro reference clip, chosen per clip by ear (team decision 2026-10-01). Short vowels may instead be a team member's recordings voice-converted with ElevenLabs (team decision 2026-10-08). The Blend It words AM, TUB and YAM and the short /i/ were made with ElevenLabs Text to Speech after the Kokoro and earlier synthetic takes failed review (team decisions 2026-10-11 and 2026-10-12). These are Free-plan clips: non-commercial use only, and the app credits "elevenlabs.io" in the Parent Zone.

2. No phoneme model shall be synthesized by feeding single letters into raw text-to-speech. Continuous sounds shall utilize phoneme IPA tokens or phonetic interjections (e.g., *"Mmm!"*); short sounds shall utilize human phoneme recordings.
3. Every shipped audio clip shall pass three strict verification gates:
   - **Gate 1 (Internal Acoustic Audit):** Zero audible trailing schwa, clipped at ≤ 800 ms (continuous) or ≤ 250 ms (stops), SNR ≥ 25 dB.
   - **Gate 2 (Vosk Technical Verification):** Phoneme audio tested against Vosk recognizer to verify target acceptance without tripping foil tokens.
   - **Gate 3 (Expert Pedagogical Audit):** Independent audit by ≥ 3 certified reading teachers rating the clip as "Pure".
4. A structured JSON Asset Manifest for each release (`docs/audio-release/<date>/manifest.json`) shall document each clip's ID, source, tool version, voice, duration, license, and gate verification timestamps.

[NFR-ASR-01] Vosk Recognition Agreement, Fairness & Error Categorization (Refactored · P1)

- **Requirement:**
  1. The Vosk speech recognition engine shall achieve ≥ 80% agreement with certified teacher perceptual judgments on vocal productions from Grade 1 Filipino learners.
  2. **False-Reject Bound:** The engine shall maintain a false-reject rate of ≤ 15% on teacher-rated correct attempts to prevent learner frustration.
  3. **Acoustic Adaptation:** The recognition grammar shall accommodate Philippine English phonological variations (e.g., lax/tense vowel substitutions, /p/-/f/ and /b/-/v/ developmental variations).
  4. Rejections shall be categorized into discrete diagnostic error types: `LETTER_NAME`, `ADDED_VOWEL`, `SUBSTITUTION`, or `UNKNOWN`.

[NFR-PERF-01] End-to-End Latency Thresholds (Refactored · P1)

- **Requirement:**
  1. **Say It Latency:** Measured from the acoustic termination of child speech (`speech_end`) to feedback rendering (`feedback_shown`), latency shall be ≤ 0.5 s at the 90th percentile (P90).
  2. **Find It Latency:** Measured from card tap (`find_tap`) to visual highlight/sound cue, latency shall be ≤ 0.3 s at P90.
  3. **Tap-to-Mic Latency:** Microphone state transition from Idle to Listening shall occur in ≤ 100 ms.

3.3.2 Pedagogical Usability & Accessibility

[NFR-IND-01] Independent Learner Usability & Dual-Coding (New · P1)

- **Requirement:**
  1. Grade 1 children must be able to navigate and complete all lessons independently without an adult present.
  2. All prompts must be spoken (dual-coded with clear vector icons). No activity shall require reading written instructions.
  3. Onboarding and first-time interactions shall feature an animated demonstration by Lily the Mascot.
  4. **Idle Re-Prompt:** If no user input is registered for 10 s, the mascot shall execute an idle bounce and replay the instructional carrier prompt.

[NFR-ACC-01] Visual Articulation Guides & Sound Captions (New · P2)

- **Validation Finding:** Accessibility checklist revealed a 60% score on hearing accommodations (`ACC-06`) due to absence of visual phonetic cues.
- **Requirement:**
  1. Hear It and Say It shall display a synchronized `ArticulationCue` component illustrating proper mouth, lip, and teeth placement for each phoneme. **Amended (team decision 2026-10-11, card 28c):** the cue is shown in Say It, beside the microphone from the second miss (FR-03.4); Hear It shows none, because a second picture during the modeling sequence splits a pre-reader's attention between the mouth and the letter.
  2. On-screen text captions representing the pure phonetic sound (e.g., *"mmm"*, *"sss"*, *"aaa"*) shall accompany audio playback to assist hearing-impaired learners. **Amended (team decision 2026-10-10, card 28):** captions are not shown in the child view, because pre-readers cannot read the carrier sentences and the text pulls their eyes from the letter; the mouth-shape cue (item 1) is the visual support. The caption logic stays in the code for a possible accessibility option.

[NFR-ACC-02] Pediatric Touch Target Ergonomics (Refactored · P2)

- **Validation Finding:** One of five caregivers reported difficulty for learners with motor challenges, linked to tight padding around the corner menu toggles (`ACC-07`, 4 of 5 Yes).
- **Requirement:** Every clickable or interactive element across child-facing and adult-facing screens—including corner back arrows, settings icons, and ear buttons—shall enforce a minimum touch target bounding box of 64 × 64 dp with at least 8 dp separation from adjacent targets.

[NFR-SES-01] Developmental Session Pacing (New · P2)

- **Requirement:** To prevent cognitive exhaustion in 6-year-old learners, **[Planned — Round 2 Scope]** the application shall gracefully terminate a learning session at the first completed node after approximately 12 minutes of continuous gameplay, celebrating earned stars and returning to the profile screen.

3.3.3 Reliability, Offline Security & Architecture

[NFR-REL-01] Transactional Offline Persistence (Refactored · P1)

- **Requirement:** The application shall commit state changes to SQLite/Room immediately upon the completion of every sub-level interaction. An abrupt OS process termination or battery failure shall result in zero loss of completed progress.

[NFR-SEC-01] Complete Local Data Privacy

- **Requirement:** 100% of learner telemetry, vocal audio buffers, and profile records shall remain strictly within the device's private app sandbox (`/data/user/0/com.playit.app/`). The application shall contain zero network permissions (`android.permission.INTERNET` omitted from `AndroidManifest.xml`).